Janvi P. Madhani

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Profile

Rising fifth-year Physics and Astronomy graduate student at Johns Hopkins University with demonstrated experience in instrumentation, space telescope survey design, data analysis, super-computing, and hydrodynamic simulations. Strongly committed to more diversity in STEM through long-haul mentoring projects in collaboration with Baltimore Area and NYC public universities. Interested in developing theoretical models, numerical simulations, and pattern recognition projects related to Cosmology, Galaxy Evolution, Earth & Atmospheric physics, and Response to Climate Change.

Research Appointments

since March 2023

LSST/NRAO Graduate Research Assistant | City University of New York, NYC College of Technology, NY
advised by Dr. Charlotte Welker.
responsibilities: research, undergraduate mentoring, outreach
research topics: planes of satellite galaxies, pattern detection in large-scale structures, cosmological hydrodynamic simulations, data analysis

since Aug 2020

Graduate Research Assistant | Johns Hopkins University, MD
advised by Dr. Charlotte Welker and Dr. Colin Norman
responsibilities: research, teaching
research topics: hydrodynamic simulations, pattern detection in large-scale structures, JWST and CLASS (CMB) survey design, preparation and/or calibration

June 2021- Jan 2023

Graduate Research Assistant | Space Telescope Science Institute, MD
advised by Dr. Charlotte Welker and Dr. Susan Kassin.
responsibilities: research, teaching
research topics: pattern detection in large-scale cosmic structures, JWST galaxy survey design and preparation

Education

since Aug 2020

Johns Hopkins University, Baltimore, Maryland, USA
Ph.D. Candidate in Physics & Astronomy
advisor: C. Welker, C. Norman

2015 - 2019

University of Pittsburgh, Pittsburgh, Pennsylvania, USA
B.S. in Physics & Astronomy, *magna cum laude*

Research Projects

Since June 2021	**Planes of Satellite Galaxies** | Johns Hopkins University, advisor: C. Welker. *Skills: fluid dynamics modeling and programming, data analysis, pattern recognition, supercomputing* • Resolved a long-standing tension between satellite galaxies' kinematics and the standard model of cosmology, using cosmological, hydrodynamic simulations. • developed infrastructural and analysis software in Fortran and Python to extract dark matter and gas filaments, identify planar structures among galaxies and filaments and detect alignments patterns in cosmological simulations. • Developed a method to adapt and calibrate a structure finding algorithm to identification of local gas streams around galaxies as well as large scale cosmic filaments. • Conducted an in-depth analysis that evidenced strong correlations between satellite dynamics and gas dynamics on larger scale (vorticity) (1st author publication in prep.)
Since June 2021	**Galaxies in Cosmic Filaments at $z = 3$ with JWST** | Space Telescope Science Institute, advisors: C. Welker, S. Kassin. *Skills: space telescope survey planning, spectroscopy, calibration* • Core co-I of a successful JWST Cycle 2 proposal to detect the influence of cosmic filaments on the spin galaxies at $z \approx 3$ with deep spectroscopy. (PI: Kassin) • Demonstrated the possibility to detect cosmic filaments on CEERS galaxy data, developed, tested and calibrated a filament detection pipeline. • Estimated the minimal specifications of the planned spectroscopic survey for optimal detection of galaxy spin alignments with filaments.
Aug 2020 - May 2021	**Cosmology Large Angular Scale Surveyor (CLASS)** | Johns Hopkins University, advisor: T. Marriage. *Skills: atmospheric noise analysis, instrumentation, signal processing, data reduction* • Conducted data selection and analysis with CLASS, a large angular scale cosmic microwave background (CMB) experiment searching for gravitational B-modes • Developed the post-demodulated cuts pipeline to clean data for CMB mapping. • Characterized the atmospheric polarization signal at large angular scales. • Tested and assembled cryogenic instrumentation.
Sept 2019 - Jul 2020	**Allegheny Observatory Parallax Catalogue** | University of Pittsburgh, advisor: D. Turnshek. *Skills: database creation and management* • Automated the creation of catalogues of parallaxes for the Allegheny Observatory, cross-matched them with the Yale Parallax Catalogue, SIMBAD and AO historic photographic plates. • Created a searchable database of stars (specific to the observed Allegheny Observatory parallaxes).

Testing a Theoretical Cosmological Model | University of Pittsburgh, advisor: A. Kosowsky.

Jan 2019 - Apr 2020

Skills: Analytical cosmology modeling

• Investigated the effects of an exotic cosmological model on the power spectrum of the Cosmic Microwave Background using both analytical and computational models.

NASA Eclipse Ballooning Project | University of Pittsburgh, advisor: D. Turnshek.

Sept 2016 - Nov 2019

Skills: programming, atmosphere analysis, instrumentation, mechanical engineering

• Studied "shadow bands", the bands of light and dark racing across the surface of the earth right before totality, with a high-altitude balloon during the 2017 total solar eclipse.

• Designed and created a lab shadow band simulator as well as balloon and ground-based photo-diode circuits.

•Analyzed eclipse data in search of shadow bands in the upper atmosphere. (led to 1st author publication)

Searching for Fast Radio Bursts with the ACT Telescope | University of Pittsburgh, advisor: A. Kosowsky.

Jan 2018 - May 2019

Skills: programming, structure identification

• Automated the identification of Fast Radio Bursts (FRBs) candidated in the Atacama Cosmology Telescope (ACT) data.

• Successfully developed a pipeline to differentiate cosmic rays from glitch data.

Teaching and Mentoring Experience

CUNY City Tech, since 2023

LSST Research Mentor

Activities: co-advising on research projects, lead of collaborative code development, graduate application counselling, conference chaperon

• Undergraduate Students: Lianys Feliciano, Daniel Gallego

• Graduate Students: Manuel Ramirez, Sneha Nair, Sophia Miskiewicz

STScI/ Johns Hopkins 2022

Astro Scholars Instructor

This NSF program offers a week-long boot camp in research, programming, and astrophysics, followed by tailored mentoring and monthly meetings for college students from underrepresented groups in the Baltimore and D.C. area.

• Participated in group meetings and mentoring

• Conducted reviews and interviews as a member of the admission committee

• Developed and taught the week-long introductory Python course (10 students).

Johns Hopkins 2021-2022

Research Mentorship

Activities: co-advising on astronomy topics, technical support, co-working sessions

• High School Student, Catherine Franklin

• Undergraduate Student, Angela Wroblewski

Johns Hopkins since 2020

JHU Teaching Assistant
• Stellar Structure and Evolution (*Grad, Fall 2023*), taught by Dr. Kevin Schlaufman.
• General Physics for Physical Sciences (*Summer 2023*), taught by Dr. Reid Mumford.
• General Physics for Bio Majors I (*Fall 2020*), taught by Dr. Tobias Marriage.
• General Physics Lab I (*Fall 2020*), taught by Dr. Reid Mumford.

U. of Pittsburgh 2018 - 2020

Undergraduate Teaching
•Undergraduate Tutor for the Physics Department
•Undergraduate Teaching Assistant for Quantum Mechanics I (2018)

COMPUTATIONAL SKILLS

Programming

Computer Languages
• Fluent: Python, bash, git, Matlab.
• Familiar: `FORTRAN`, C

Analysis

Software & Packages
• `numpy`, `scipy`, `matplotlib`, `astropy`, `pandas`, LaTeX, Microsoft Office, `RAMSES`, `DisPerSE`, `SUNSET`, `MUSIC`, SolidWorks

GRANTS AND FUNDING AWARDS

Travel Grants

IAU S379: Dynamical Masses of Local Group Galaxies
PI, Awarded by Leibniz Institut fur Astrophysik March 2023, €1100

AAS FAMOUS Travel Grant
PI, Awarded January 2022, $1000

Research

JWST 4291: Galaxy angular momentum alignment with filaments at z $\sim$ 3: The effect of large scale structure on galaxies
Co-Investigator, Cycle 2 - 2023 (PI:Kassin)

Space Telescope Science Institute Director's Discretionary Research Fund
Awarded 2021 and 2022, $9000 for Summer and $60,000 for a full year
Co-I (PI: Kassin, Welker)

NASA - Pennsylvania Space Grant Consortium Scholarship
PI, Awarded 2017, 2018, 2019, Cumulative total of $15,000

OUTREACH

Jan 17, 2020

Panel Speaker for CUWiP, *University of Pittsburgh*
• Spoke on panels to share my experiences as an early career researcher.

Oct 2016 2017, 2018, 2019

Allegheny Observatory Open House, *University of Pittsburgh*
• Volunteered at the annual Allegheny Observatory open house giving talks some years and doing physics demonstrations other years for the general public.

Jan 30, 2019

Science Research Panel Speaker – Internship Week, *University of Pittsburgh*

March 2018 **Public Lecture - Allegheny Observatory Open House**, *University of Pittsburgh*

Professional Service

May 2-4, 2022 **Congressional Visits Day**, *Washington D.C.*
• Met with members of Congress to advocate for science policy and research funding on behalf of the AAS.

In the Press

June 14, 2022 **Press Conference at AAS 240**, *Pasadena, CA*
• Recording can be found here!

Aug 21, 2017 **Live Radio Interview with P.J. Maloney of KQV AM 1410**, *Pittsburgh*

Aug 16, 2017 **Shadow Bandits Ready for Eclipse Day**, *University of Pittsburgh*
• Click to read!

July 15, 2017 **It's up, up, and away...**, *University of Pittsburgh*
• Click to read!

Refereed Publications

1. **Madhani, J. M.**, Chu, G. E., Gomez, C.V., Bartel, S., Clark, R.J., Coban, L. W., Hartman, M., Potosky, E.M., Rao, S.M., Bonato, M., Turnshek, D. A., *"Observation of Eclipse Shadow Bands Using High Altitude Balloon and Ground-Based Photodiode Arrays,"* **JASTP**, 2020, 211, 105420.

Invited Talks

Apr 19, 2023 **"The Formation of Planes of Satellite Galaxies"** | University of Connecticut Galaxy Seminar (*virtual*).

March 16, 2018 **Plenary Talk: "Eclipse Shadow Bands"** | Duquesne University's Summer Research Symposium (*Pittsburgh, PA*)

Sept 22, 2017 **Plenary Talk: "Observations of Eclipse Shadow Bands"** | Allegheny Observatory Open House (*Pittsburgh, PA*)

Contributed Talks & Lectures

July 24, 2023 **"The Formation of Planes of Satellite Galaxies"** | Galactic Frontiers: Dwarf Galaxies in the Local Volume and Beyond (*Center for Computational Astrophysics, NYC*)

March 24, 2023 **"The Formation of Planes of Satellite Galaxies"** | IAU S379: Dynamical Masses of Local Group Galaxies (*Potsdam, Germany*)

Feb 6, 2023 **"Conflict Resolved: Planes of Satellites Found in Simulations"** | IAU S377: Early Disk-Galaxy Formation from JWST to the Milky Way (*Kuala Lumpur, Malaysia*)

Sept 23, 2022 **"Planes of Satellite Galaxies"** | Galaxy Group Meeting at the Center for Computational Astrophysics (*virtual*)

June 13, 2022 **"The Formation of Planes of Satellite Galaxies"** | 240th Meeting of the AAS (*Pasadena, CA*)

Posters

July 29-Aug 2, 2024	**Planes of Satellite Galaxies Trace Environmental Whirling Gas Streams in the Cosmic Web** | Small Galaxies, Cosmic Questions (*Durham, UK*)
March 20-24, 2023	**The Formation of Planes of Satellite Galaxies** | IAU S379: Dynamical Masses of Local Group Galaxies (*Potsdam, Germany*) • Won "Most Innovative and Creative Poster" for which I was awarded a contributed talk slot.
May 23-27, 2022	**Planes of Satellite Galaxies in New Horizon** | IAA-CSIC Severo Ochoa Advanced School on Galaxy Evolution (*Granada, Spain*)
Jan 4, 2019	**Observations of Eclipse Shadow Bands** | 235th Meeting of the AAS (*Honolulu, HI*)
Oct 13, 2016	**NASA Eclipse Ballooning Project** | White House Frontiers Astronomy Night (*Pittsburgh, PA*)

Additional Information

Languages	English (native), Gujarati (fluent), Hindi (fluent), French (conversational)

Journal abbreviations used

JASTP | *Journal of Atmospheric and Solar Terrestrial Physics*

This CV was last updated on August 29, 2024.